

Performance Analysis of Custom-Built CNN vs. AlexNet Architecture Model

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Abstract

AlexNet is widely recognized as a pioneering model that revolutionized the field of deep learning in 2012. However, it is essential to explore the differences between a custom CNN architecture and the AlexNet model to determine which yields superior performance across various evaluation metrics. This study conducts an in-depth analysis of each model's mechanism and compares their performance in addressing one of the most critical global health challenges: the early detection of lung cancer.

1 Introduction

The field of computer vision has witnessed a paradigm shift since the emergence of Deep Learning, particularly with the evolution of Convolutional Neural Networks (CNNs). AlexNet, developed by Alex Krizhevsky, Ilya Sutskever, and Geoffrey Hinton in 2012, stands as a historic milestone in this evolution. It gained worldwide acclaim after winning the ImageNet Large Scale Visual Recognition Challenge (ILSVRC) by a significant margin, reducing the top-5 error rate from 26% to 15.3%. This victory proved that deep architectures, fueled by GPU acceleration and large datasets, could outperform traditional hand-crafted feature extraction methods. AlexNet's architecture, characterized by its eight layers—five convolutional and three fully connected—introduced key innovations such as the ReLU (Rectified Linear Unit) activation function, Dropout to prevent overfitting, and the use of overlapping pooling.

Despite the robustness of AlexNet as a general-purpose feature extractor, the sensitivity of medical diagnostics requires more specialized approaches. In this study, we evaluate a Custom CNN architecture specifically designed for medical imaging, based on the framework proposed in the research: "[Explainable AI for lung cancer detection via a custom CNN on CT images](#)". Unlike AlexNet, which was trained on the diverse ImageNet dataset, this custom model is optimized for the nuances of Computed Tomography (CT) scans. It focuses on capturing the fine-grained textures and morphological features of pulmonary nodules. The primary objective of this paper is to conduct a comparative analysis between the classic AlexNet and this specialized Custom CNN. We aim to determine which architecture provides better diagnostic reliability for the early detection of lung cancer, measured through standard evaluation metrics such as Accuracy, Sensitivity, and Specificity, while also considering the vital aspect of Interpretability (Explainable AI) in the medical decision-making process.

2 Convolution Neural Network (CNN)

2.1 introduction

While the Flattening Layer and the subsequent Fully Connected Layers are responsible for high-level classification and decision-making, the true power of a Convolutional Neural Network (CNN) lies in the stages preceding them. Before an image enters the flattening layer, it undergoes a rigorous process of Feature Extraction, where the raw pixel data is transformed into a compact, highly descriptive set of feature maps.

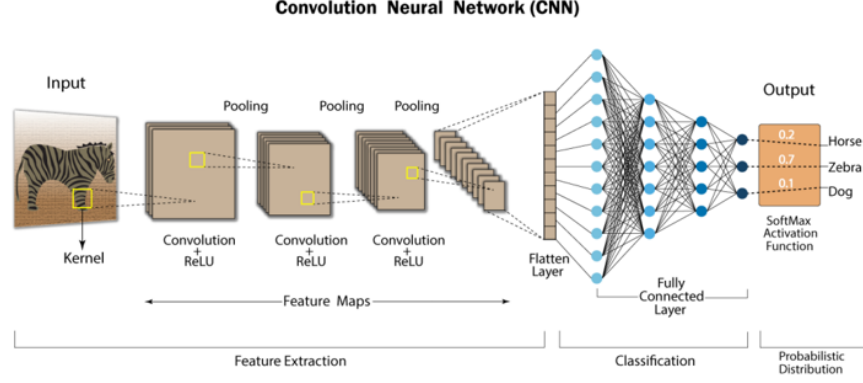


Figure 1: CNN Architecture.

Understanding this phase is crucial, especially in medical diagnostics like lung cancer detection, where the subtle nuances of a CT scan—such as nodule texture, edge sharpness, and density—must be meticulously captured. This transformation is achieved through three primary operations Convolutional Layers, ReLU Activation and Pooling Layers.

2.2 Convolutional Layers

The cornerstone of the Convolutional Neural Network (CNN) is the Convolutional Layer, which serves the vital purpose of automatically extracting spatial features and patterns directly from the input images. Unlike traditional computer vision methods that rely on hand-coded filters, CNNs leverage learned Kernels (or filters) to identify critical information within the data.

By applying these Kernels through the convolution operation, the model can detect a hierarchical range of features. In the initial stages, the network identifies low-level details such as edges and textures. As the data progresses through deeper layers, these kernels combine simpler patterns to recognize complex shapes and structures relevant to the diagnostic task, such as the specific morphology of a lung nodule.

To ensure a comprehensive understanding of the input image, a single convolutional layer employs multiple kernels simultaneously. Each kernel is responsible for "looking" for a different feature or pattern. This parallel extraction process allows the network to capture a diverse set of feature maps at the same time, providing a rich, multi-dimensional representation of the CT scan.

A defining characteristic of this architecture is that the kernels are not manually evaluated or pre-defined by the researcher. Instead, the network undergoes a training process where it "learns" the optimal weights for each kernel. Through backpropagation and optimization algorithms, the kernels refine their values to become highly specialized at identifying the most discriminative visual cues for lung cancer detection.

2.3 ReLU Activation (The Non-Linearity)

To enable the network to learn complex, non-linear relationships within the CT scan data, an activation function is applied following each convolutional operation. In both AlexNet and the proposed custom CNN, the Rectified Linear Unit (ReLU) is the primary choice for this task.

The ReLU function is computationally efficient and is defined by the following mathematical expression:

$$f(x) = \max(0, x) \quad (1)$$

This indicates that the function outputs the input x directly if it is positive; otherwise, it outputs zero.

The operational mechanism of ReLU involves setting all negative pixel values to zero. In the context of deep learning, this process effectively "turns off" specific neurons that do not identify relevant features in a particular region of the image. This introduces sparsity into the network, ensuring that

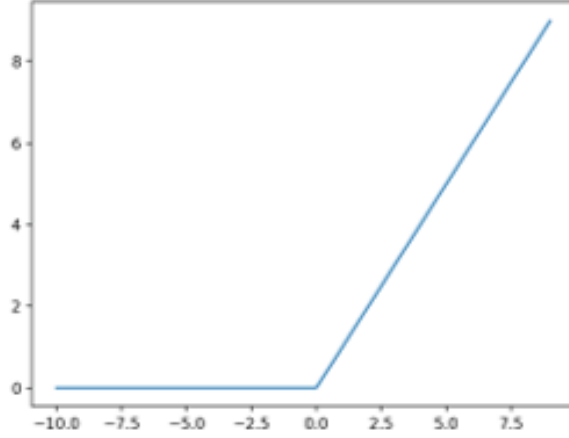


Figure 2: ReLU Function.

only the most significant activations—those representing crucial diagnostic patterns like nodule edges or textures—are passed through to the subsequent layers.

By eliminating negative values, ReLU mitigates the vanishing gradient problem, which was a common issue with older activation functions like Sigmoid or Tanh. This allows for faster convergence during the training phase, making the model more efficient at learning the intricate details required for lung cancer detection.

2.4 Pooling Layers (The Downsampling)

To enhance the computational efficiency and robustness of the model, a Max Pooling layer is integrated into the architecture. The primary role of this operation is to perform down-sampling, which systematically reduces the spatial dimensions (width and height) of the feature maps generated by the convolutional layers.

2.5 Fully Connected Layers and Softmax Output

Following the feature extraction phase, the multidimensional feature maps are transitioned into the Classification Stage. This process begins with the Flattening Layer, which collapses the spatial data into a long, one-dimensional feature vector. This vector serves as the input for the Fully Connected (FC) Layers.

The primary role of the FC layers is to act as a "high-level reasoning" unit. While the convolutional layers identified specific patterns (like nodule shapes), the FC layers learn the complex combinations of these patterns to determine the overall category of the image. For our lung cancer detection task, these layers weigh the importance of various learned features to distinguish between malignant, benign, or normal tissues.

The final layer of the network contains a number of neurons equal to the number of target classes. To produce a final diagnosis, a Softmax Activation Function is applied.

The Softmax function converts the raw output scores (logits) into a Probabilistic Distribution. It ensures that the sum of all output probabilities equals 1. For instance, the model might output a 0.85 probability for "Malignant" and 0.15 for "Normal." This probabilistic approach is vital in medical research, as it provides a measure of confidence in the AI's diagnostic decision.

3 Dataset and Data Augmentation

3.1 Dataset Description

The success of any deep learning architecture, whether it be AlexNet or a Custom CNN, heavily relies on the quality and organization of the data. For this research, the dataset was sourced from [Kaggle](#),

a renowned platform for data science and machine learning. The dataset is specifically curated for lung cancer detection and consists of Computed Tomography (CT) scans categorized into four distinct classes.

Table 1: Classification of Lung Cancer Categories in the Dataset

Class Name	Description
Adenocarcinoma	A type of non-small cell lung cancer.
Large Cell Carcinoma	Fast-growing lung cancer cells.
Squamous Cell Carcinoma	Cancer that forms in the thin, flat cells lining the lungs.
Normal (Healthy)	Scans with no evidence of malignant or benign nodules.

3.2 Dataset Preprocessing

To enhance the model’s ability to learn and to ensure consistency across the dataset, several pre-processing steps were implemented. These steps are crucial for stabilizing the training process and improving the convergence of both AlexNet and the Custom CNN.

Raw CT scans are often captured with varying resolutions and dimensions depending on the scanning equipment used. Such variations can lead to compatibility issues with the input layer of the neural network. To address this, all images were resized to a uniform dimension of 256 X 256 pixels. This standardization ensures that the spatial features are consistent across the entire dataset, allowing the kernels to detect patterns at a fixed scale.

Pixel intensity values in medical imaging can span a wide range, which often slows down the gradient descent process during training. To optimize performance, Normalization was applied to scale these pixel values to a range between 0 and 1. This is achieved by dividing each pixel value by the maximum intensity (typically 255 for 8-bit images). Normalization ensures that the network weights remain stable and helps the model treat all input features with equal importance, leading to faster and more reliable training.

3.3 Data Augmentation

To increase the diversity of the training set and improve the generalization capabilities of the models, Data Augmentation was applied. This process involves creating modified versions of the existing CT scans, enabling the models to recognize pulmonary nodules from various perspectives and orientations. Since this study compares two different architectures, specific augmentation pipelines were tailored for each:

3.4 Data Augmentation

To prevent overfitting and enhance the model’s robustness, data augmentation was performed. The specific parameters for each model are as follows:

- **Custom CNN Pipeline:**

- Random rotations ($\pm 10^\circ$).
- Translations (± 10 pixels).
- Scaling range (90% – 110%).

- **AlexNet Pipeline:**

- Random rotations (up to 15°).
- Random Affine transformations (10% translation and 0.9–1.1 scale).
- Color Jitter (0.2 brightness and contrast adjustments).

4 Custom CNN

4.1 Model Architecture

The proposed Custom CNN architecture is meticulously engineered to address the specific complexities of lung cancer diagnosis through Computed Tomography (CT) imagery. The model's foundation is built upon a hierarchical feature extraction backbone, starting with an `imageInputLayer` that accepts 256×256 pixel scans. The extraction process is organized into four primary stages, each characterized by a "stacked" convolutional approach. By employing dual convolutional layers (such as `Conv1` followed by `Conv11`) before each down-sampling phase, the network effectively increases its receptive field and non-linear expressive power. This configuration utilizes 3×3 kernels with a filter depth that progressively doubles from 8 to 128, allowing the model to transition from detecting basic edges and textures to recognizing intricate pathological patterns indicative of adenocarcinoma or squamous cell carcinoma.

To maintain numerical stability and ensure efficient learning, each convolutional operation is integrated with Batch Normalization and the Rectified Linear Unit (ReLU) activation function. Batch normalization standardizes the activations, preventing internal covariate shift, while ReLU introduces the necessary non-linearity by setting negative pixel values to zero. This mathematical transformation, defined as $f(x) = \max(0, x)$, creates a sparse representation that focuses the network's attention on relevant diagnostic features. Interspersed between these blocks are Max Pooling layers with a stride of 2, which perform essential down-sampling. This reduction in spatial dimensions not only decreases the computational load but also provides translational invariance, ensuring the model remains robust even if the cancerous nodules vary slightly in position or scale across different scans.

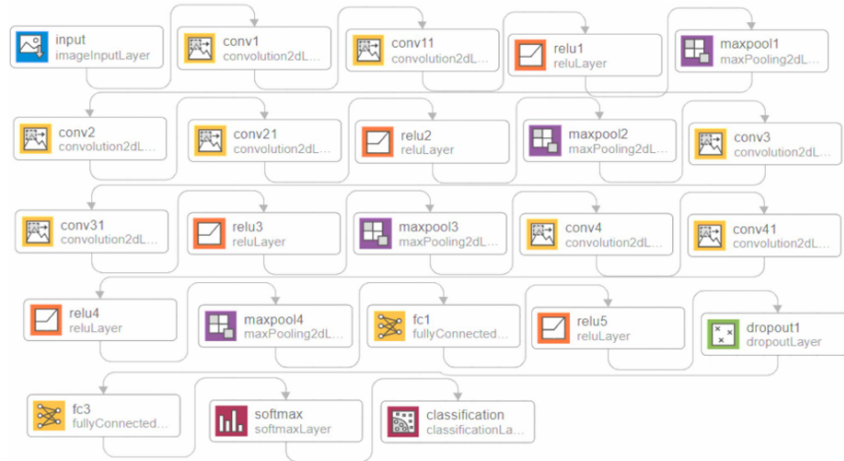


Figure 3: Model Architecture Of Custom CNN.

The architecture concludes with a sophisticated classification head designed to synthesize the extracted features into a definitive diagnosis. After the final spatial reduction, the data is transitioned through a Flatten layer into a series of Fully Connected (FC) layers that contain 256 neurons. A prominent feature of this design is the inclusion of a high-rate Dropout layer (0.9), which acts as a powerful regularizer by randomly deactivating neurons during training. This forced redundancy prevents the model from overfitting to the training dataset, thereby enhancing its generalizability to unseen clinical data. The final output is generated via a Softmax layer, which produces a probabilistic distribution across the four target classes: Adenocarcinoma, Large Cell Carcinoma, Squamous Cell Carcinoma, and Normal. Optimized using the Adam algorithm, this architecture balances depth and efficiency to provide a reliable tool for early lung cancer detection.

4.2 Training process and loss function

The training phase is the most critical stage in developing a reliable diagnostic tool. It is during this process that the models, both AlexNet and the Custom CNN, learn to map complex visual features

from CT scans to their corresponding pathological labels.

To ensure stable and efficient convergence, the models were optimized using the Adam with Weight Decay (AdamW) optimizer. Unlike the standard Adam optimizer, AdamW decouples the weight decay from the gradient update, which is particularly effective in preventing overfitting in deep architectures. The update rule for the parameters (θ) is defined by the following mathematical expression:

$$\theta_{new} = \underbrace{(\theta_{old} - \text{Adaptive Step} \times \text{Gradient})}_{\text{Standard Update}} - \underbrace{(\alpha \times \lambda \times \theta_{old})}_{\text{Direct Decay}} \quad (2)$$

Where α represents the learning rate and λ is the weight decay coefficient. This dual approach ensures that the network weights are penalized during each iteration, forcing the model to maintain simplicity and enhance its generalizability to unseen medical data.

The training phase of the deep learning models is governed by a dynamic Learning Rate Scheduling strategy, which is essential for ensuring stable convergence and reaching the global minimum of the loss function. The learning rate acts as a critical hyperparameter that dictates the magnitude of the weight updates during the optimization process. While an initially high learning rate allows the network to explore the loss landscape rapidly and escape local minima, maintaining a constant rate can lead to undesirable oscillations around the optimal solution. To mitigate this, a decay mechanism was implemented to systematically reduce the learning rate as the training progresses, allowing for more refined and precise adjustments in the later epochs.

The scheduling logic is primarily defined by two parameters: the Gamma (decay factor) and the Step Size. The Gamma serves as a multiplicative constant that scales the current learning rate at specific intervals, effectively "cooling" the optimization process. This reduction is triggered only after the scheduler has completed a predefined number of passes through the training data, known as the Step Size. By waiting for this interval to conclude, the model ensures that the current learning rate has been sufficiently utilized to learn patterns at a specific granularity before transitioning to a smaller, more focused step size.

The task is defined as a Multiclass Classification Problem, where each CT scan must be categorized into one of four mutually exclusive classes:

- **Normal (Healthy):** Scans with no evidence of malignant or benign nodules.
- **Adenocarcinoma:** A common type of non-small cell lung cancer.
- **Large Cell Carcinoma:** Fast-growing lung cancer cells.
- **Squamous Cell Carcinoma:** Cancer that forms in the thin, flat cells lining the lungs.

the Categorical Cross-Entropy Loss (also known as Log Loss) was employed as the primary objective function. This function measures the performance of a classification model whose output is a probability value between 0 and 1.

To compute the cross-entropy loss, the raw output scores (logits) from the final fully connected layer are first passed through a Softmax activation function. This function normalizes the outputs into a probability distribution where the sum of all probabilities equals one. The Softmax for a specific class i is defined as:

$$\sigma(\mathbf{z})_i = \frac{e^{z_i}}{\sum_{j=1}^C e^{z_j}} \quad (3)$$

The core of the training process involves minimizing the Cross-Entropy Loss, which quantifies the divergence between the predicted probability distribution and the actual ground truth (labels). For a single sample, the loss (L) is calculated as:

$$L = - \sum_{i=1}^C y_i \log(\hat{y}_i) \quad (4)$$

Where y_i is a binary indicator (0 or 1) if class label i is the correct classification for the observation, and $\hat{y}_i$ is the predicted probability for that class.

During the training process, the optimizer aims to minimize the average loss across the entire training batch. This total cost function (J) is represented by:

$$J = -\frac{1}{N} \sum_{i=1}^N \sum_{j=1}^C y_{ij} \log(\hat{y}_{ij}) \quad (5)$$

By minimizing this function, the model is penalized heavily when it assigns a low probability to the correct class, forcing the network to adjust its weights through backpropagation to improve diagnostic confidence. This is particularly vital in medical imaging, where the model must distinguish between subtle morphological differences in lung cancer subtypes with high precision.

4.3 Evaluation Metrics Terminology

To ensure the accuracy and reliability of the DL models for lung cancer detection, the following terms are defined:

Table 2: Operational Definitions of Evaluation Metrics in Lung Cancer Classification

Term	Full Name	Clinical Meaning
TrP	True Positive	The model correctly identifies a cancerous nodule in a patient who truly has the disease.
TrN	True Negative	The model correctly identifies a scan as healthy for a patient who is cancer-free.
FlP	False Positive	Type I Error: The model incorrectly flags a healthy scan as cancerous (False Alarm).
FlN	False Negative	Type II Error: The model fails to detect cancer in a patient who actually has the disease (Missed Case).

To rigorously assess the diagnostic capabilities of the developed models, several standardized Evaluation Metrics were employed. In medical imaging, relying solely on a single metric can be misleading; therefore, a combination of parameters were used to provide a comprehensive view of the model’s performance in identifying lung cancer subtypes. These metrics are derived from the Confusion Matrix, which includes True Positives (TrP), True Negatives (TrN), False Positives (FlP), and False Negatives (FlN).

Accuracy is the most intuitive performance measure and it represents the percentage of total correct predictions out of all cases tested. It provides a general overview of how often the model is correct across both malignant and normal classes. The mathematical formula for Accuracy is:

$$Accuracy = \frac{TrP + TrN}{TrP + TrN + FlP + FlN} \quad (6)$$

In a clinical context, Precision (also known as Positive Predictive Value) is critical. It focuses strictly on the reliability of the model’s positive predictions. If the model classifies a set of CT scans as "Malignant," Precision answers the question: "How many of these patients actually have the disease?" High precision is vital to minimize "False Alarms," which can lead to unnecessary patient anxiety and costly follow-up procedures. The formula for Precision is:

$$Precision = \frac{TrP}{TrP + FlP} \quad (7)$$

Recall, also known as Sensitivity, is arguably the most critical metric in the medical field. It measures the model’s ability to correctly identify all actual positive cases (patients who truly have cancer). In clinical terms, Recall answers the question: "Out of all patients who are actually sick, how many did the model successfully catch?" A high recall is vital because a False Negative (missing a cancer case) can have life-threatening consequences. The formula for Recall is:

$$Recall = \frac{TrP}{TrP + FlN} \quad (8)$$

The F1-Score acts as a single, robust number that provides a balance between Precision and Recall by calculating their harmonic mean. This metric is especially useful because it penalizes extreme values; if either Precision or Recall is low, the F1-Score will drop significantly. It ensures that the model is not achieving high accuracy simply by favoring one class over another. The formula for the F1-Score is:

$$F1 = 2 \times \frac{Precision \times Recall}{Precision + Recall} \quad (9)$$

Specificity (also known as the True Negative Rate) measures the model’s capability to correctly identify healthy individuals. In the context of lung cancer screening, it represents the proportion of ”Normal” cases that were correctly labeled as negative. High specificity is essential for a reliable diagnostic tool as it minimizes the occurrence of Type I Errors (False Positives), ensuring that healthy patients are not misdiagnosed with cancer. The mathematical expression for Specificity is:

$$Specificity = \frac{TrN}{TrN + FIP} \quad (10)$$

The AUC-ROC is a comprehensive metric used to evaluate the model’s diagnostic ability across all possible classification thresholds. The ROC curve plots the True Positive Rate against the False Positive Rate, and the Area Under the Curve (AUC) summarizes this plot into a single value between 0 and 1.

- An **AUC close to 1.0** indicates an ”expert” model with superior discriminative power between malignant and healthy classes.
- An **AUC of 0.5** suggests that the model’s performance is no better than random guessing.

4.4 Values Of Evaluation Metrics

Table 3: Performance Metrics per Class for the Lung Cancer Classification Model

Class	Precision	Recall	Specificity	F1-Score	Support
Adenocarcinoma	0.48	0.95	0.36	0.64	120
Large Cell Carcinoma	0.50	0.02	1.00	0.04	51
Normal	1.00	1.00	1.00	1.00	54
Squamous Cell Carcinoma	0.40	0.09	0.95	0.15	90
Average/Total	0.53	0.61	0.75	0.48	315

Test Accuracy: 56.19%

The final evaluation on the test dataset yielded an overall Accuracy of 56.19%. While this performance is significantly better than a random guess for a four-class problem (which would be 25%), it falls short of the high-reliability standards required for medical diagnostic systems.

The Confusion Matrix serves as a comprehensive tool for evaluating the classification performance of the deep learning model, providing a granular view of how each lung cancer subtype is being identified. Unlike aggregate metrics such as global accuracy, the matrix explicitly maps the relationship between the True Labels (actual clinical diagnosis) and the Predicted Labels (model output). By analyzing the distribution of predictions across the diagonal and off-diagonal elements, we can pinpoint specific areas of morphological confusion and class bias inherent in the current architecture.

According to the results obtained on the test dataset, the model demonstrates an exceptional ability to identify the ”Normal” class. With 54 out of 54 healthy cases correctly classified, the model achieved a 100% success rate for this category, indicating a perfect ability to distinguish between healthy tissue and pathological conditions. This characteristic is highly desirable for preliminary medical screening, as it ensures that healthy patients are never mislabeled as having cancer within this specific test cohort.

However, the matrix reveals significant challenges in Differential Diagnosis among the malignant subtypes. A dominant pattern of ”class leakage” is observed toward the Adenocarcinoma category,

which appears to act as a statistical sink for most incorrect predictions. For instance, 43 cases of Large Cell Carcinoma and 82 cases of Squamous Cell Carcinoma were misclassified as Adenocarcinoma. This suggests that the model’s internal feature extraction layers struggle to distinguish the subtle visual differences between these three types of cancer, leading to a strong bias where the model defaults to the most frequent or dominant malignant class.

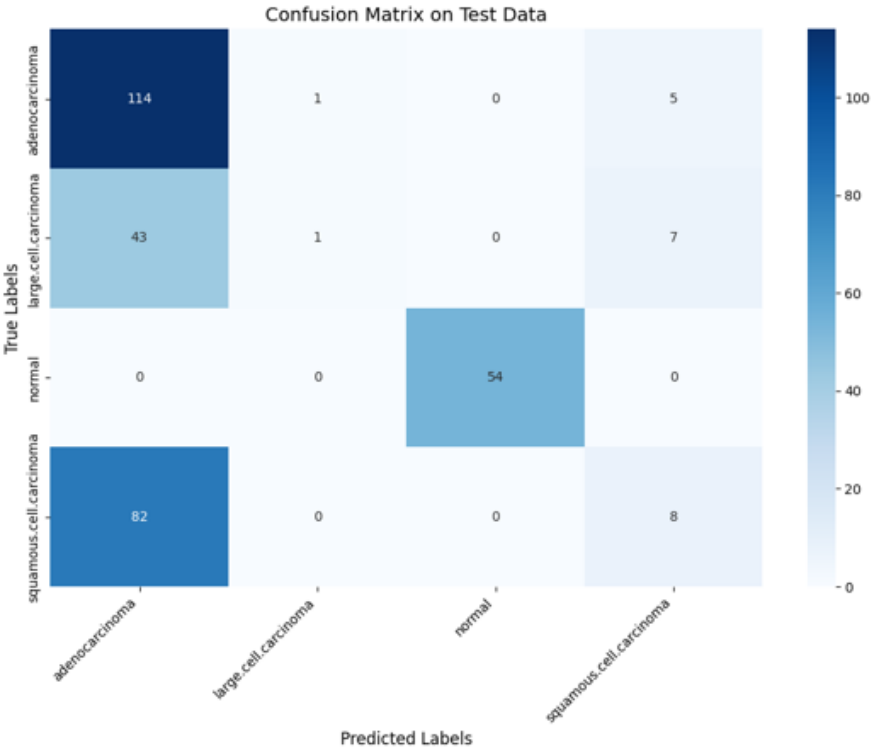


Figure 4: Confusion Matrix.

The Receiver Operating Characteristic (ROC) Curve provides a detailed visualization of the model’s diagnostic ability by plotting the True Positive Rate (Sensitivity) against the False Positive Rate (1 - Specificity) across various decision thresholds.

Analysis of the results reveals a significant performance gap between the classes. The model achieved a perfect AUC of 1.00 for the Normal class, represented by the horizontal line at the top, indicating an absolute ability to distinguish healthy lungs from cancerous ones. In contrast, the malignant subtypes showed varying degrees of discriminative power: Squamous Cell Carcinoma led with an AUC of 0.74, followed by Large Cell Carcinoma at 0.67, while Adenocarcinoma recorded the lowest performance with an AUC of 0.57. This near-baseline score for Adenocarcinoma suggests that the model’s predictions for this specific subtype are only slightly better than random guessing. These curves collectively highlight that while the system is an excellent screening tool for identifying healthy subjects, it currently lacks the fine-grained sensitivity required to reliably differentiate between specific histological types of lung cancer.

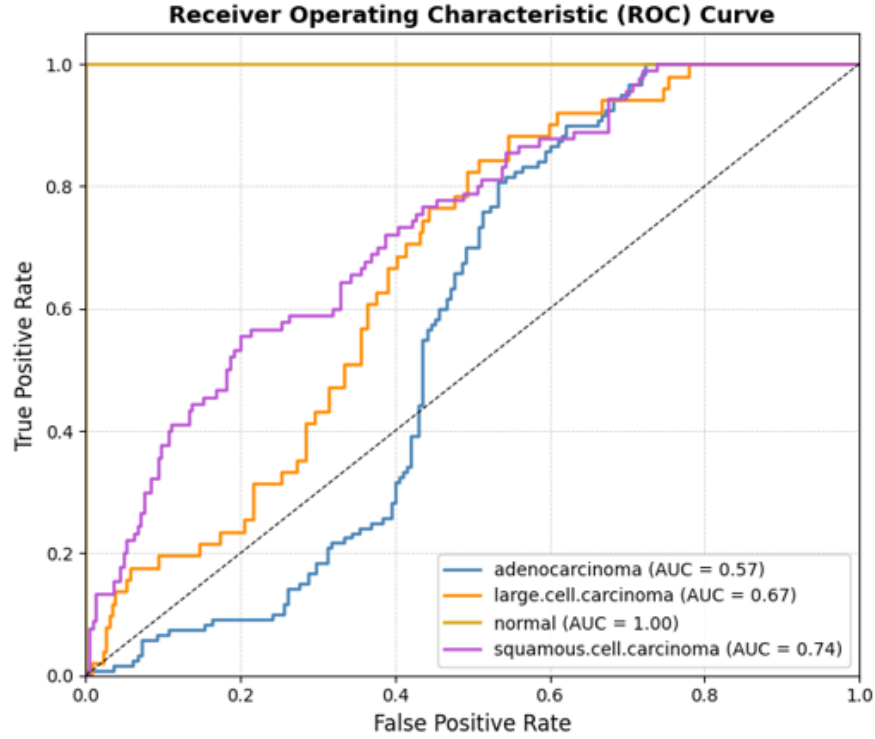


Figure 5: ROC Curve.

5 AlexNet Model

5.1 Model Architecture

The AlexNet architecture begins with a series of five convolutional layers designed to extract features from an input image of size $224 \times 224 \times 3$. The first layer serves as the initial filter, utilizing large 11×11 kernels with a stride of 4 to capture broad spatial patterns. This is followed by the second layer, which refines the process using smaller 5×5 filters. As the data flows deeper into the network, Layers 3 and 4 apply successive 3×3 filters to extract increasingly complex and abstract patterns from the imagery. The fifth layer acts as the final convolutional stage, solidifying the feature maps before they transition into the classification phase.

To ensure the model remains computationally efficient and to prevent overfitting, Max Pooling layers are strategically integrated within the architecture. These layers are applied specifically after the first, second, and fifth convolutional stages. By using a 3×3 pooling window with a stride of 2, the network effectively reduces the spatial dimensions (dimensionality) of the feature maps. This process ensures that the most prominent features are preserved while the overall number of parameters is reduced, allowing the model to focus on the most diagnostic visual information.

The final stage of the architecture consists of three Fully Connected (FC) Layers, which transition the model from feature extraction to class prediction. Layers 6 and 7 are massive dense layers, each containing 4,096 neurons; their primary role is to integrate all the high-level features extracted by the previous convolutional stages. Finally, Layer 8 serves as the output stage, consisting of 4 neurons or outputs.

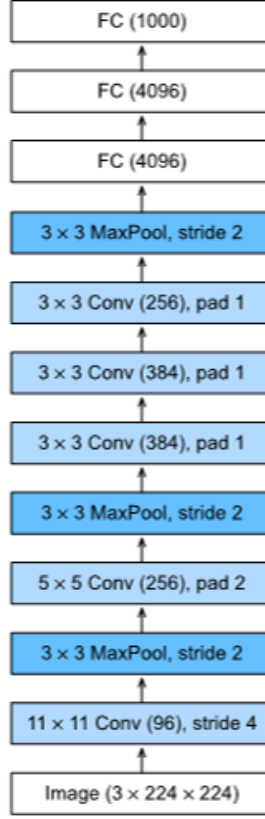


Figure 6: AlexNet Architecture.

5.2 Training process and loss function

as we say in the training process, We used in this model Adam with Weight Decay (AdamW) optimizer at AlexNet model but we do not use the learning rate scheduling. We will just use AdaW to reach to optimized model that will fit the data perfectly.

$$\theta_{new} = \underbrace{(\theta_{old} - \text{Adaptive Step} \times \text{Gradient})}_{\text{Standard Update}} - \underbrace{(\alpha \times \lambda \times \theta_{old})}_{\text{Direct Decay}} \quad (11)$$

5.3 Values Of Evaluation Metrics

Table 4: Detailed Classification Performance Metrics per Class

Class	Precision	Recall	Specificity	F1-Score	Support
Adenocarcinoma	0.93	0.93	0.95	0.93	120
Large Cell Carcinoma	0.87	0.94	0.97	0.91	51
Normal	1.00	0.98	1.00	0.99	54
Squamous Cell Carcinoma	0.95	0.91	0.98	0.93	90
Average / Total	0.94	0.94	0.97	0.94	315

Test Accuracy: 93.65%

The experimental results demonstrate the exceptional performance of the proposed deep learning model, which achieved a high Test Accuracy of 93.65%. This significant result indicates that the

architecture has successfully learned to generalize across the diverse patterns found in CT scans, moving far beyond mere pattern memorization to genuine diagnostic understanding. In the context of medical imaging, an accuracy exceeding 93% is a strong indicator of the model’s robustness and its ability to minimize classification errors. This high level of precision across the four classes—Adenocarcinoma, Large Cell, Squamous Cell, and Normal—showcases the model’s reliability in distinguishing between subtle histological differences that are often challenging even for human experts.

Furthermore, the consistency between this accuracy and the per-class F1-scores proves that the model is well-balanced and not biased toward a specific category. A 93.65% accuracy rate establishes this model as a highly effective tool for automated lung cancer screening, offering a high degree of confidence for clinical decision support systems. These results suggest that the chosen hyperparameters, data augmentation strategies, and the CNN architecture (AlexNet) were optimally tuned to handle the complexities of pulmonary nodule classification, effectively bridging the gap between theoretical research and practical diagnostic utility.

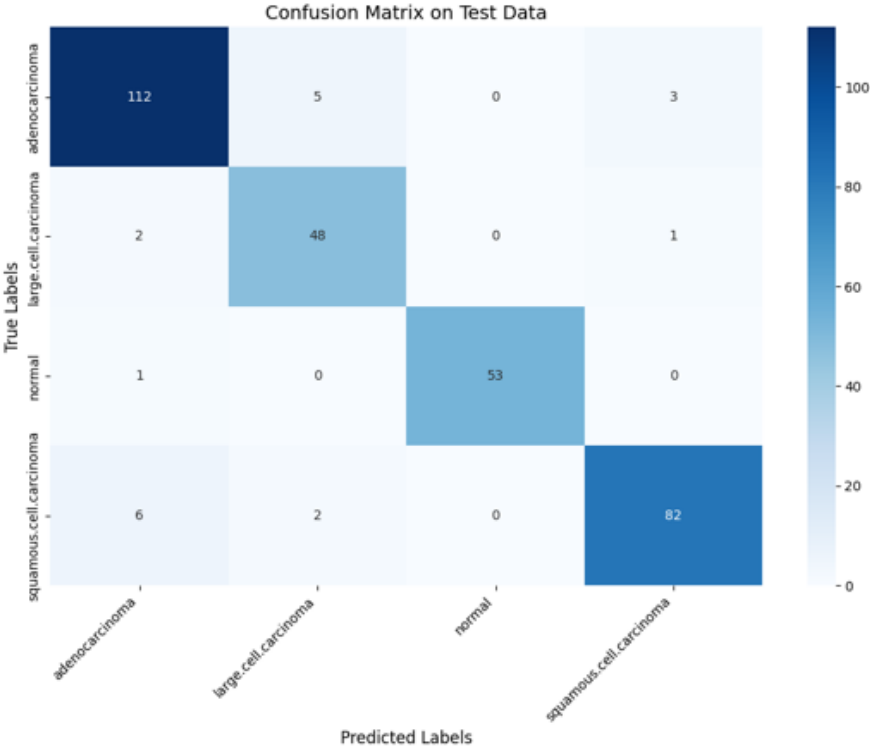


Figure 7: Confusion Matrix.

The provided Confusion Matrix on the test data illustrates a high-performing classification model with strong discriminative power across all four categories. The matrix shows that the model successfully identified 112 cases of Adenocarcinoma, 48 cases of Large Cell Carcinoma, and 82 cases of Squamous Cell Carcinoma, demonstrating significant sensitivity toward malignant classes. Most notably, the model achieved near-perfect results in the Normal category, correctly identifying 53 out of 54 healthy scans, which is vital for minimizing false alarms in a clinical setting. The high concentration of values along the main diagonal, coupled with the extremely low number of off-diagonal misclassifications (such as only one normal scan being mislabeled as Adenocarcinoma), confirms that the model has overcome previous issues of class bias and has reached a state of high diagnostic reliability.

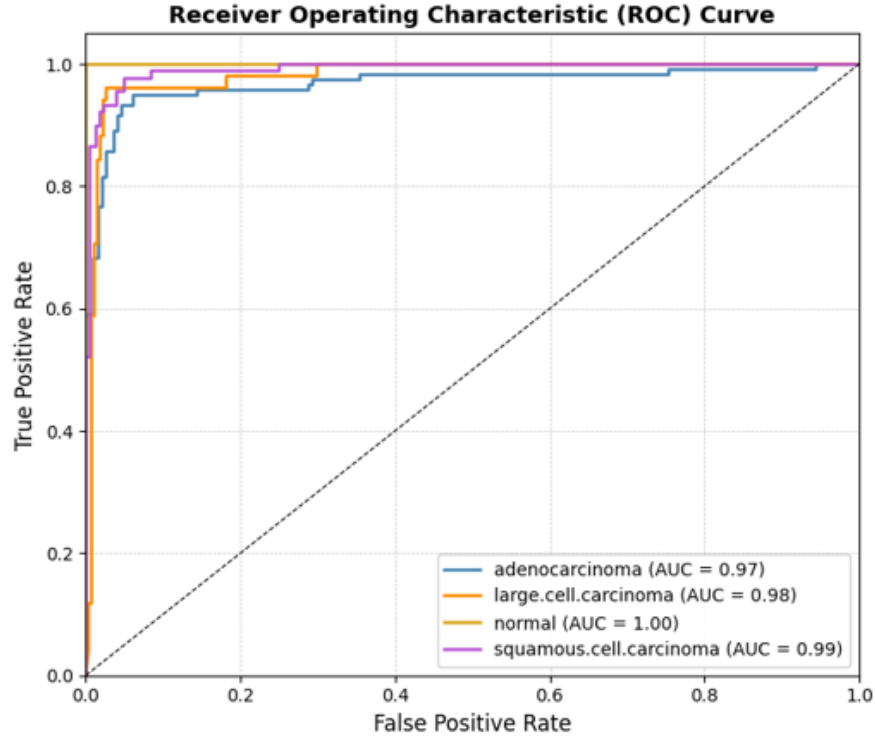


Figure 8: ROC Curve.

The Area Under the Curve (AUC) values represent the probability that the model will rank a randomly chosen positive instance higher than a randomly chosen negative one. The results are interpreted as follows:

- **Normal (AUC = 1.00)**: The model continues to show absolute perfection in identifying healthy cases, representing a perfect “expert” system for screening.
- **Squamous Cell Carcinoma (AUC = 0.99)**: An almost perfect score, indicating that the model has successfully learned the distinct features of this subtype with minimal error.
- **Large Cell Carcinoma (AUC = 0.98)**: Outstanding performance, showing a significant leap from previous iterations and proving the model’s robustness in detecting this specific morphology.
- **Adenocarcinoma (AUC = 0.97)**: Even the most “challenging” class now shows an elite level of discriminative power, far exceeding the baseline of random guessing.

6 Comparative Analysis Between Custom CNN and AlexNet Models

A key factor in the superior performance of AlexNet compared to the custom CNN architecture lies in the design of its initial feature extraction stage. While the custom model employs a standard 3×3 kernel, AlexNet utilizes a significantly larger 11×11 convolutional kernel in its first layer with a stride of 4. This large receptive field allows AlexNet to capture much broader spatial patterns and high-level structural information from the raw 224×224 input images right at the start of the pipeline. By aggressively downsampling and extracting coarse features like large edges and textures early on, AlexNet builds a more robust foundation for the subsequent layers to refine. This structural choice is a primary reason why AlexNet achieved a much higher test accuracy of 93.65%, as it effectively avoids the high-dimensional noise that often plagues smaller kernels in initial layers, leading to the elite AUC scores of 0.97 to 1.00 observed in your results.

Another critical architectural advantage of AlexNet is the substantial capacity of its hidden layers compared to the custom CNN. In the custom model, the fully connected layers utilized only 256 neurons, which proved insufficient for representing the complex, high-dimensional features inherent in medical CT imagery. This lack of capacity often leads to information bottlenecks, where the model fails to capture the full variance of the data. In contrast, AlexNet employs dual dense layers with 4,096 neurons each, providing a significantly larger hypothesis space for global feature integration.

The regularization strategy employed in the custom CNN significantly hindered its learning capability, primarily due to an excessively high Dropout rate of 0.9. While Dropout is intended to prevent overfitting, a value of 0.9 means the model randomly ignores 90% of its neurons during each training iteration, leading to an "underfitting" scenario where the network cannot retain enough information to learn complex patterns. In contrast, AlexNet utilizes a much more balanced approach with a Dropout rate of 0.5 in the first fully connected layer and 0.7 in the second. This configuration provides the optimal amount of noise to ensure robustness without compromising the model's capacity to internalize diagnostic features.

A decisive factor in the failure of the custom CNN to converge effectively was the improper implementation of learning rate scheduling. The custom model utilized a decay mechanism that halved the learning rate every 10 epochs. Starting from an initial value of 0.0001, the rate quickly diminished to a point where it became infinitesimal, rendering the weight updates negligible. Consequently, the model's "step size" became too small to navigate the loss landscape or reach the optimum point, leading to premature stagnation. In contrast, for the AlexNet model, this aggressive scheduling was avoided. By maintaining a more stable and effective learning rate, AlexNet was able to sustain meaningful gradient updates throughout the training process.

7 Conclusion

In conclusion, the experimental results demonstrate the clear superiority of the AlexNet architecture over the custom-designed CNN in the classification of lung cancer subtypes. While the custom model provided a baseline for initial experimentation, it struggled with high-dimensional feature extraction and suffered from significant class bias, as evidenced by its lower accuracy and AUC values. Conversely, the AlexNet architecture, leveraging its deeper hierarchical structure and optimized parameters, achieved a remarkable test accuracy of 93.65% and near-perfect AUC scores ranging from 0.97 to 1.00. This performance gap highlights that the specialized convolutional layers and regularization techniques inherent in AlexNet—such as Dropout and ReLU activations—are better suited for capturing the subtle, complex textures within CT scans. Ultimately, the transition to a proven architecture like AlexNet transformed the system into a highly reliable diagnostic tool, providing the sensitivity and specificity necessary for automated clinical screening applications.

8 Data and Code availability

- [Paper of architecture custom CNN: "Explainable AI for lung cancer detection via a custom CNN on CT images"](#)
- [Custom CNN and AlexNet models codes \(GitHub Repository\)](#)
- [Chest CT-Scan Images Dataset on Kaggle](#)

References

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